

Technical Fit Checklist

SAN TECH · Umujoyanama w'Ubuhinzi · EOI: AI-Enabled Agricultural Advisory Solution Partner (C4IR)

Live product: <https://santech-agri-advisor.vercel.app> · API docs: <https://santech-agri-api.onrender.com/docs>

Contact: *[Name]* · *[email]* · *[phone]*

A. Access to our solution for the benchmark

Statement	Answer	Evidence
Our solution is available behind one secure API endpoint	Yes	POST /ask, protected by an X-API-Key header. Interactive docs at /docs.
It answers farmer questions in English and Kinyarwanda, including follow-ups	Yes	Language is detected or set; session_id keeps the conversation for follow-up questions.
It provides a readiness check	Yes	GET /health returns ready: true/false and the database status.
It records which version answered, and how fast	Yes	Each answer returns model, system_version, prompt_version and latency_ms; GET /version; every request is logged.
Voice (optional): Kinyarwanda audio or text in, audio or text out	Yes	POST /transcribe (speech-to-text) and POST /speak (text-to-speech).
The model and harness (prompts, retrieval, knowledge base, guardrails, orchestration) are all behind the API	Yes	Channel front-ends (web, USSD, SMS) call the same API.
We can take part in onboarding on 9–14 October	<i>[Yes]</i>	Contact: <i>[name, email, phone]</i>
We can add C4IR's local data for the final evaluation	Yes	Documents and data are added with scripts/ingest.py, without code changes; the same tests are then re-run.

B. Hosting and data in Rwanda

Statement	Answer	Evidence
Today's pilot hosting	Cloud pilot	Application: Vercel and Render. Database: Supabase PostgreSQL (EU, Ireland). Language model: Google Gemini API.
We will host the solution and data in Rwanda for production	Yes, committed	Standard PostgreSQL and a stateless Python API move to a Rwanda-based data centre (<i>[provider to be confirmed]</i>) without code changes.
We can use a model hosted in Rwanda or an approved provider	Yes	The model is set by configuration (LLM_BASE_URL, LLM_MODEL); any OpenAI-compatible endpoint, including a self-hosted open model, works.
C4IR data will not be processed on free consumer tiers	Yes, committed	Free tiers are used only for this public pilot with test data.
Compliance with Law No. 058/2021 on personal data protection	Yes, by design	No farmer phone numbers stored (pseudonymous ID for SMS); photos not stored; no names on dashboards; officer records refuse phone and ID numbers; database tables locked from public access (row-level security).

We will sign C4IR's data sharing agreement	<i>[Yes]</i>	
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C. Openness, documentation and transfer

Statement	Answer	Evidence
Source code is open or available to C4IR	<i>[Open source / On request]</i>	https://github.com/YVES10BOBO/santech-agri-advisor <i>[public or private; licence]</i>
Built with open-source technology	Yes	Python, FastAPI, PostgreSQL, pgvector, Next.js; the model provider is swappable.
Documentation	Yes	README (setup and what was built), docs/api.md (all endpoints), docs/architecture.md (design, security, roadmap), interactive /docs.
Automated tests	Yes	28 automated tests; a 50-question bilingual benchmark script with a scorecard.
Knowledge transfer to C4IR and government teams	Yes	<i>[Handover sessions, training for engineers, documentation in English and Kinyarwanda]</i>
Ownership and IP	To be agreed at RFP stage	As stated by C4IR.

Known limits, stated openly

- Kinyarwanda wording and agronomic numbers still need review by a native speaker and an agronomist before production.
- Pilot hosting is outside Rwanda (see B).
- Logins and roles for dashboards, WhatsApp, IVR, weather alerts and live input and market data are on the roadmap.